// IIIT HYDERABAD * PGEE 2026 * PAPER I * GENERAL APTITUDE

The complete student guide // Fira Code Edition

HOW TO PREPARE

APTITUDE

FOR PGEE 2026

01 QUANTITATIVE

02 VERBAL

03 ANALYTICAL

04 SPATIAL

Exam: 2nd May 2026 // Paper I: 90 min // CBT // All India

pgadmissions.iiit.ac.in

>_ Designed for PGEE 2026 aspirants // Fira Code Edition

Quant * Verbal * Analytical * Spatial

01 $ Quantitative Aptitude

$ QUANTITATIVE APTITUDE

Numbers * Arithmetic * Algebra * Geometry * DI // 30-35%

Numbers | Percentage | Profit-Loss | TSD | Work | Interest | Algebra

Quantitative Aptitude is the largest and most formula-intensive domain. Build strong arithmetic fluency first — it unlocks 70% of all quant questions. Every formula below has appeared in PGEE or GATE GA recently.

[01]Number System

-> Types: Natural, Whole, Integer, Rational, Irrational, Prime, Composite
-> LCM & HCF — applied in grouping, timing, and tile-covering problems
-> Divisibility rules: 2, 3, 4, 5, 6, 7, 8, 9, 11 — memorize every one
-> Remainders, cyclicity of units digit (1^n=1, 2^4k=6, etc.)
-> Surds, indices (laws of exponents), BODMAS order of operations

[02]Percentage, Ratio & Proportion

-> Percentage increase / decrease and successive % change formula
-> Reverse percentage: find original value from final changed value
-> Ratio simplification, comparison; direct & inverse proportion
-> Partnership: capital x time = profit ratio

[03]Profit, Loss & Discount

-> CP, SP, MP relationships | Profit% = (Profit / CP) x 100
-> Successive discounts: equivalent = 100 - (1-d1/100)(1-d2/100)x100
-> Dishonest dealings: faulty weights, false marking scenarios

[04]Time, Speed & Distance

-> Speed = Distance / Time | km/h x 5/18 = m/s
-> Relative speed: same direction = difference, opposite = sum
-> Train crossing: pole (length/speed), platform (length+platform)/speed
-> Boats & streams: upstream = boat_speed - stream, downstream = sum
-> Average speed for equal distances = 2*S1*S2 / (S1+S2)

[05]Time & Work / Pipes & Cisterns

-> Work rate = 1/days per day | A+B together = 1/A + 1/B
-> Fractional work in x days = x * (1/total_days)
-> Pipes: filling rates (+) and emptying rates (-) combined

[06]Simple & Compound Interest

-> SI = P*R*T / 100 | Amount = P + SI
-> CI = P*(1 + R/100)^T - P | half-yearly: use R/2 and T*2
-> CI - SI for 2 years shortcut = P*(R/100)^2

[07]Averages & Alligation

-> Average = Sum / Count | Weighted average = sum(wi*xi) / sum(wi)
-> Adding / removing a member — how it shifts the group average
-> Alligation cross method: mix two prices/strengths for target

[08]Algebra & Sequences

-> Linear & simultaneous equations | Quadratic and discriminant D=b^2-4ac
-> AP: nth = a+(n-1)d | Sum = n/2*(first+last)
-> GP: nth = a*r^(n-1) | Sum = a*(r^n - 1)/(r-1)
-> Key identities: (a+b)^2, (a-b)^2, a^2-b^2, (a+b)^3 — all must be instant

[09]Geometry & Mensuration

-> Triangles: Pythagoras, similarity, congruence, area = (1/2)*b*h
-> Circles: chord, tangent, sector area = (theta/360)*pi*r^2
-> 2D areas: rectangle, parallelogram, trapezoid, rhombus, circle
-> 3D: cube, cuboid, cylinder, cone, sphere — volume AND surface area

[10]Permutation, Combination & Probability

-> FCP: multiply independent stage choices
-> nPr = n!/(n-r)! | nCr = n! / (r! * (n-r)!)
-> Circular arrangements: (n-1)! for n distinct objects
-> P(E) = Favourable / Total | P(A and B) = P(A)*P(B) if independent

[11]Data Interpretation (DI)

-> Bar chart: read heights carefully, compute % change between bars
-> Pie chart: sector angle = (value/total) * 360 degrees
-> Line graph: slope = rate of change; steeper = faster change
-> Table DI: row/column operations, % of total calculations
-> Mixed DI: two linked charts — always cross-reference before answering

$ Quant — Top Tips

-> Vedic maths shortcuts: fast % calculation saves 15+ seconds per question.
-> Always convert units FIRST in TSD — km/h vs m/s is the most common trap.
-> In DI: read chart title + axis labels before ANY question — saves re-reading.
-> AP/GP: know both nth-term AND sum formula — both appear in one question.
-> Algebra shortcuts: memorize the 5 key identities — they appear every year.

02 % Verbal Aptitude

% VERBAL APTITUDE

Read fast, comprehend deeply, reason from text // 25-28%

Reading Comprehension | Para Jumbles | Para Completion | Critical Reasoning

PGEE Verbal focuses entirely on reading, comprehension, and text-based reasoning. There is NO grammar or vocabulary testing. Every question is passage-based or argument-based. Strong readers can answer these in under 45 seconds each, making Verbal the fastest domain when prepared well.

[01]Reading Comprehension (RC)

-> Passage length: 200-450 words with 4-5 questions per passage
-> Main idea / Central theme — what is the passage PRIMARILY about?
-> Specific detail — directly stated fact, locate in the passage text
-> Inference — what can be concluded but is NOT directly stated?
-> Author's tone — critical / appreciative / neutral / satirical / analytical
-> Title selection — must cover the FULL passage, not just one paragraph
-> Meaning-in-context — what does the underlined word mean in this passage?
-> Strategy: read the questions FIRST, then skim the passage for answers
-> Eliminate extreme / absolute choices — RC answers avoid 'always' or 'never'
-> The correct answer is ALWAYS supported by the passage — no outside knowledge

[02]Para Jumbles (Sentence Rearrangement)

-> 4-6 scrambled sentences — rearrange into a logical, coherent paragraph
-> Step 1: Find the MANDATORY FIRST sentence — no pronoun / connector at start
-> Step 2: Find the MANDATORY LAST sentence — conclusion, result, or summary
-> Step 3: Pair sentences using pronoun -> antecedent links (he, she, it, this)
-> Step 4: Use logical connectors — however / therefore / moreover / but / so
-> Step 5: Verify the final paragraph reads naturally from start to finish
-> Common trap: two sentences may link either way — use full paragraph context

[03]Para Completion

-> Incomplete paragraph given — choose the BEST opening or closing sentence
-> Opening completion: must introduce the topic, no pronoun without prior context
-> Closing completion: must conclude logically, often echoes the opening theme
-> Eliminate any choice that introduces a new unrelated idea
-> Correct sentence must match the tone of the passage (formal, casual, critical)

[04]Critical Reasoning — Argument Based

-> Short argument (4-6 lines) + a question about its logical structure
-> Strengthen: choose evidence that makes the conclusion MORE likely true
-> Weaken: choose evidence that makes the conclusion LESS likely true
-> Assumption: what hidden fact MUST be true for the argument to work?
-> The assumption = the invisible bridge between evidence and conclusion
-> Inference: what MUST be true based ONLY on the information given?
-> FIJ format — correctly classify each statement: Fact / Inference / Judgment

[05]Statement — Conclusion & Inference

-> One or more statements given -> determine if conclusion logically follows
-> Implicit: not directly stated but logically follows from the information
-> Explicit: directly stated or very clearly derivable without assumptions
-> Trap: a conclusion TRUE in reality but NOT supported by the given statements
-> Always base your answer ONLY on the statements, never on general knowledge

[06]Logical Connectors & Flow Questions

-> Choose the correct connector to fill a blank in a passage
-> Cause-effect: because / therefore / consequently / as a result
-> Contrast: however / although / despite / nevertheless / on the other hand
-> Addition: furthermore / moreover / in addition / also / besides
-> Wrong connector changes the logical relationship — read the full sentence

% Verbal — Top Tips

-> RC strategy: questions first, then skim — saves 40 seconds per passage.
-> Para Jumbles: mandatory first sentence (no pronoun/connector) is your anchor.
-> Critical Reasoning: assumption = hidden bridge between evidence and conclusion.
-> FIJ: Judgment = someone's opinion, Fact = verifiable, Inference = derived.
-> Never use outside knowledge in RC — the answer must come from the passage only.

03 ? Analytical Reasoning

? ANALYTICAL REASONING

Highest ROI domain — logic and patterns reward daily practice // 25-28%

Series | Coding-Decoding | Blood Relations | Directions | Arrangements

Analytical Reasoning delivers the best marks-per-hour return of all four domains. Questions follow strict logical patterns — once you know the approach for each type, they become the most reliable marks in your paper.

[01]Number & Letter Series

-> Difference series: +3, +5, +7 (arithmetic) or x2, x3 (ratio pattern)
-> Square / cube series: 1, 4, 9, 16... or 1, 8, 27, 64...
-> Alternating series: two separate series interleaved into one
-> Mixed operations: +2, x3, +2, x3 — alternating operation pattern
-> Letter series: A, C, F, J (gaps +1,+2,+3 — find position skip rule)
-> Wrong term: find the one number that breaks the established pattern

[02]Coding & Decoding

-> Letter-shift: each letter shifted by fixed n — A+3=D (Caesar cipher)
-> Reverse alphabet: A<->Z, B<->Y — first position = last position
-> Number-position: A=1,B=2... or reversed A=26,B=25...
-> Symbol substitution: decode by finding the transformation pattern
-> Conditional coding: vowels coded differently from consonants

[03]Blood Relations

-> Draw a family tree — always start from the person mentioned first
-> Multi-step: 'A's mother's brother's son' — map every link on the tree
-> Gender detection from relationship clues when gender is ambiguous
-> Coded relations: +, -, x symbols each represent a relationship type

[04]Directions & Distance

-> 8 compass points: N NE E SE S SW W NW — know angles between them
-> Left turn = anticlockwise 90 | Right = clockwise 90 | About = 180
-> Net displacement: use Pythagoras for right-angle (N then E) paths
-> Shadows: morning -> West | evening -> East

[05]Seating Arrangements

-> Linear row: single row, double-row facing, mixed position conditions
-> Circular table: clockwise vs anticlockwise changes 'immediate left'
-> ALWAYS draw a blank diagram first — never work only in your head
-> Start with the most constrained person (most rules) — place them first
-> Verify every single condition after completing the arrangement

[06]Scheduling, Floor & Box Puzzles

-> Floor puzzles: N people on N floors with above/below/adjacent conditions
-> Box puzzles: items in ordered stack — build an elimination grid
-> Day/month scheduling: who did what on which day — build a matrix
-> Comparison chains: A>B, B>C -> deduce full rank order from all clues

[07]Multi-Attribute Logic Puzzles

-> Match N people to N values across M attributes: city, job, hobby, color
-> Build elimination grid: rows = people, columns = attributes
-> Fill definite clues first, then use negatives to eliminate options
-> Cross-reference: if A must be in Mumbai, no one else can be in Mumbai
-> These are hardest but most rewarding — practice 5+ puzzles per week

[08]Syllogisms

-> Format: All/Some/No A is B + statement 2 -> valid conclusion?
-> Draw a Venn diagram for EACH statement before testing any conclusion
-> Definite conclusions vs possible conclusions — exam tests both
-> Key rule: All A is B + All B is C -> All A is C (always valid)
-> Some rules: Some A is B + All B is C -> Some A is C (valid)

[09]Clocks & Calendars

-> Hour hand: 0.5 deg/min | Minute hand: 6 deg/min
-> Angle = |30H - 5.5M| degrees (H=hours, M=minutes past the hour)
-> Hands overlap: 11 times in 12 hours, 22 times in 24 hours
-> Leap year: divisible by 4 AND (not century OR divisible by 400)
-> Odd days method: anchor on known day, count forward/backward

[10]Data Sufficiency & Critical Deduction

-> Data Sufficiency: is the given data enough to answer uniquely?
-> Never solve the full problem — only check if it IS solvable
-> Two statements: A alone / B alone / Both / Neither sufficient
-> Cause and Effect: distinguish correlation from causation in statements

? Analytical — Top Tips

-> Arrangement puzzles: DRAW the diagram. Every. Single. Time. No exceptions.
-> Syllogisms: draw a Venn diagram for each statement — never guess from reading.
-> Multi-attribute puzzles: start with the person / variable with fewest options.
-> Series: check differences first, then ratios, then squares — in that order.
-> Clock angle: |30H - 5.5M| — practice until you apply it in under 20 seconds.

04 @ Spatial Aptitude

@ SPATIAL APTITUDE

Train your eyes — figures, 3D shapes, orientation // 10-15%

Mirror Images | Water Reflection | Paper Folding | Cubes & Dice

Spatial Aptitude is a pure practice-gain domain. 15 minutes of figure practice daily for 4 weeks transforms these from questions you skip into reliable easy marks. The 10-15% share becomes highly impactful.

[01]Mirror Images

-> Vertical mirror (left<->right flip): imagine mirror on the RIGHT of figure
-> Horizontal mirror (top<->bottom): imagine mirror BELOW the figure
-> Clock in mirror: 12:00 stays, 3:00 -> 9:00 (time appears reversed)
-> Text in mirror: each letter mirrored individually AND reading order reverses
-> Identify which mirror axis (vertical vs horizontal) creates the shown image

[02]Water Reflections

-> Water image = upside-down flip (top<->bottom, NOT left/right)
-> Key: Mirror = left/right flip | Water = top/bottom flip
-> Some questions combine both — identify which transformation is shown

[03]Paper Folding & Hole Punching

-> Paper folded 1-3 times, a hole punched — predict the unfolded result
-> Rule: each fold mirrors ALL existing holes across that fold line
-> Work BACKWARDS: unfold the most recent fold first, trace holes each step
-> Single punch + 2 folds = 4 total holes (2^folds when punch is single spot)

[04]Figure Series & Matrix Completion

-> Series: identify transformation rule from fig1->fig2->fig3->predict fig4
-> Common transformations: rotate 90/180 deg, reflect, add/remove elements
-> Size change: elements expand or shrink by a consistent rule each step
-> Matrix (3x3 grid): verify your rule works for BOTH rows AND columns

[05]Figure Analogy & Odd One Out

-> Analogy: first pair (fig1->fig2) shows relationship — apply same to fig3
-> Common relationships: 90 deg rotation, shading change, element addition
-> Odd one out: 5 figures — find the one that does NOT belong to the group
-> Check: number of sides, symmetry type, shading style, orientation

[06]Cubes & Dice

-> 11 valid cube nets — learn to identify which flat shapes fold into cubes
-> Painted cube: N x N x N cut into 1x1x1 cubes — count painted faces
-> Corner = 3 faces painted (always 8 corners regardless of cube size)
-> Edge = 2 faces painted | 12*(n-2) pieces for n x n x n cube
-> Standard die: opposite faces always sum to 7 — (1,6) (2,5) (3,4)
-> Two-position dice: given 2 orientations, deduce all 6 face positions

[07]3D Counting & Views

-> Count unit cubes in a stacked structure — count each column height separately
-> Front view, side view, top view — match 2D projection to 3D structure
-> Hidden cubes behind/below visible surface must STILL be counted

[08]Embedded Figures & Counting Shapes

-> Hidden figure: locate small shape inside complex figure, SAME orientation
-> The outline must use only lines already present in the complex figure
-> Counting triangles: count 1-unit + 2-unit + 3-unit + ... systematically
-> Formula: n divisions in triangle -> n(n+2)(2n+1)/8 total triangles
-> Counting rectangles in grid: C(rows+1,2) x C(cols+1,2)

@ Spatial — Top Tips

-> 15 min/day of figure practice for 4 weeks = full spatial confidence.
-> Paper folding: work BACKWARDS — unfold the last fold first every time.
-> Cube counting: count every column's height separately and sum — never guess.
-> Mirror vs Water: say it aloud — Mirror=left/right, Water=top/bottom.
-> Hidden figures: the small shape is ALWAYS there — trace it patiently.

05 = Formula Reference Sheet

Quantitative Formulas

FORMULA	EXPRESSION
Profit %	(Profit / CP) x 100
Loss %	(Loss / CP) x 100
Simple Interest	SI = (P x R x T) / 100
Compound Interest	CI = P x (1 + R/100)^T - P
Average Speed	2 x S1 x S2 / (S1 + S2) [equal distances only]
Work Rate A+B	1/A + 1/B work per unit time
AP nth term	a + (n-1) x d
AP Sum (n terms)	(n/2) x (first + last)
GP nth term	a x r^(n-1)
GP Sum (finite)	a x (r^n - 1) / (r - 1)
nCr	n! / (r! x (n-r)!)
nPr	n! / (n-r)!
Probability P(E)	Favourable outcomes / Total sample space
Cylinder Volume	pi x r^2 x h
Cone Volume	(1/3) x pi x r^2 x h
Sphere Volume	(4/3) x pi x r^3

Analytical Quick Rules

RULE	FORMULA / FACT
Clock Angle	| 30H - 5.5M | degrees (H = hours, M = minutes)
Hand Overlaps	11 times per 12 hours // 22 times per 24 hours

Circular Arrange.	(n-1)! for n distinct objects in a circle
Leap Year Test	Divisible by 4 AND (not century OR divisible by 400)
Odd Days	Ordinary year = 1 odd day | Leap year = 2 odd days

Spatial Quick Rules

RULE	FACT
Dice opposites	Always sum to 7: (1,6) (2,5) (3,4)
Valid cube nets	Exactly 11 unique flat nets fold into a valid cube
Mirror vs Water	Mirror = left/right flip | Water = top/bottom flip
Corner cubes (3-face)	Always 8 — regardless of cube size n x n x n
Edge cubes (2-face)	12 x (n-2) for n x n x n cube
Face cubes (1-face)	6 x (n-2)^2 for n x n x n cube
Triangle count	n divisions: n(n+2)(2n+1) / 8 total triangles

06 # Reference Books

Quantitative Aptitude

[01] Quantitative Aptitude for Competitive Examinations

by R.S. Aggarwal — S. Chand Publication

[QUANT] [ALL LEVELS]

The gold-standard reference for aptitude. Covers every quant topic — Number System, Profit-Loss, TSD, Algebra, Geometry, P&C; — with hundreds of graded practice questions. Start here for any quant topic.

[02] How to Prepare for Quantitative Aptitude for CAT

by Arun Sharma — McGraw Hill Education

[QUANT] [ADVANCED]

Best for DI (Data Interpretation) and advanced quant. Builds speed and accuracy with progressive difficulty levels (LOD 1, 2, 3). Excellent for students targeting 90%+ scores in quantitative sections.

[03] Fast Track Objective Arithmetic

by Rajesh Verma — Arihant Publication

[QUANT] [SHORTCUTS]

Best for Vedic maths tricks, quick-calculation shortcuts, and speed techniques. Use this alongside Aggarwal — learn the concept from Aggarwal, then pick the shortcut from Rajesh Verma.

Verbal & Reasoning

[04] A Modern Approach to Verbal & Non-Verbal Reasoning

by R.S. Aggarwal — S. Chand Publication

[VERBAL] [ANALYTICAL] [SPATIAL]

The single most complete book for Verbal (RC, Para Jumbles), Analytical (Puzzles, Arrangements, Syllogisms), and Non-Verbal (Spatial). A must-have — covers all three domains in one volume.

[05] How to Prepare for Verbal Ability and Reading Comprehension for CAT

by Arun Sharma & Meenakshi Upadhyay — McGraw Hill Education

[VERBAL] [RC] [PARA JUMBLES]

Best book for RC strategy and Para Jumbles. Teaches a systematic approach to reading comprehension with real passage practice. Highly recommended for building RC speed and accuracy.

Analytical & Logical Reasoning

[06] Analytical Reasoning

by M.K. Pandey — BSC Publishing

[ANALYTICAL] [PUZZLES] [ARRANGEMENTS]

The best dedicated book for seating arrangements, floor puzzles, scheduling, and multi-attribute logic puzzles. More comprehensive than Aggarwal for complex analytical problems. Practice every puzzle type chapter-by-chapter.

[07] How to Prepare for Logical Reasoning for CAT

by Arun Sharma — McGraw Hill Education

[ANALYTICAL] [CRITICAL REASONING]

Best for argument-based critical reasoning, data sufficiency, and advanced logic puzzles. Builds the deductive thinking needed for the toughest analytical questions in any competitive exam.

Quick Reference — Which Book for Which Topic

TOPIC	PRIMARY BOOK	SUPPLEMENT
Number System / Arithmetic	R.S. Aggarwal — Quant	Rajesh Verma (shortcuts)
Algebra / Geometry / P&C;	R.S. Aggarwal — Quant	Arun Sharma — CAT Quant
Data Interpretation	Arun Sharma — CAT Quant	R.S. Aggarwal — Quant
Reading Comprehension	Arun Sharma — Verbal & RC	R.S. Aggarwal — Verbal
Para Jumbles / Completion	Arun Sharma — Verbal & RC	R.S. Aggarwal — Verbal

Critical Reasoning	Arun Sharma — Logical Reasoning	M.K. Pandey
Seating Arrangements / Puzzles	M.K. Pandey — Analytical	R.S. Aggarwal — Reasoning
Syllogisms / Coding-Decoding	R.S. Aggarwal — Verbal & NV	M.K. Pandey
Mirror / Paper Folding / Spatial	R.S. Aggarwal — Non-Verbal	Practice sheets
Cubes / Dice / Figure Series	R.S. Aggarwal — Non-Verbal	IndiaBIX.com